

6.8 Allocation Concentration

The first step to calculating the allocation concentration of a portfolio is to define the notion of asset weight within a portfolio.

$$\text{Allocation weight}_{m \in \Pi} = \omega_m = \frac{\mathbf{x}_m}{\mathbf{x}_{\Pi}} = \frac{\mathbf{x}_m}{\sum_{m \in \Pi} \mathbf{x}_m} \quad (40)$$

This represents the percentage of each portfolio's academic capital $\mathbf{x}_{\Pi}$ that comes from each manuscript m . Each ω_m indicates the impactfulness of each asset within the portfolio.

Multiple ways to measure a portfolio's allocation concentration exist. One conventional measure is the Hefindahl-Hirschman Index (HHI), which seamlessly applies to portfolios in Liberata.

$$\text{HHI}_{\Pi} = \sum_{m \in \Pi} \omega_m^2 \quad (41)$$

From inspection, the HHI is bounded between (0,1] monotonically, with 1 being the case where there is only one manuscript's shares in the portfolio, and 0 being the case of infinitely many works with a special case of exact 0 value for a case of an empty portfolio.

Another way to measure portfolio allocation is to examine the GINI coefficient of the portfolio.

$$\text{Gini}_{\Pi} = \frac{1}{2|\Pi|} \sum_{k \in \Pi} \sum_{l \in \Pi} |w_k - w_l| \quad (42)$$

The Gini is also bounded by [0,1], where 0 would occur if all manuscripts m had the same academic capital $\mathbf{x}_m$ and 1 would occur if one manuscript accounted for the entire academic capital of the portfolio. This value varies continuously and monotonically with the inequality of weight distributions in the portfolio. Compared with the HHI, this measure will indicate precisely when there is perfect equality with the closed bound at 0, where the HHI will be affected by construction by the number of assets in the portfolio, and not only their equality. The HHI will also tend to emphasize more the large allocations, rendering diversity of small allocations more invisible in the presence of some large allocations. These small allocation diversity will be more discernible in the Gini.

A third way to measure portfolio allocation concentration is a normalized entropy measure.

$$\text{Normalized entropy} = H_{\Pi} = -\frac{1}{\log(|\Pi|)} \sum_{m \in \Pi} \omega_m \log(\omega_m) \quad (43)$$

This measure is also bounded by 1, but has the opposite interpretation, where 0 is the case of one asset having all the academic capital of the portfolio, and 1 is the case of all assets having equal same academic capital. This measure has some interesting properties regarding impact information carried within the portfolio in the unnormalized form, but in the normalized form, is an excellent measure sensitive to all scales of ω .

It is notable that these above constructions are measuring an academic portfolio's 'impact' allocation concentration, which is measured by academic capital. If instead one desired to measure effort allocation concentration within the portfolio, swap the definition of allocation weight ω_m with

$$\text{Allocation weight (shares)} = \omega'_m = \frac{s_m}{s_{\Pi}} = \frac{s_m}{\sum_{m \in \Pi} s_m} \quad (44)$$

where s_m and s_{Π} are shares held in manuscript m in portfolio Π and the latter is the simple sum of all the shares held across all manuscripts in the portfolio (shares are already normalized values that sum to unity on a manuscript).

In addition, these weights can be recalculated so that they represent not individual papers, but different tags. This would be done by batching manuscripts into primary tag objects (d_1, d_2, d_3, d_4) and computing weights by these groups, rather than by m .

$$\text{Allocation weight } (d_k | k \in \{1, 2, 3, 4\}, \mathbf{x}) = \omega_{d_k} = \frac{\mathbf{x}_{d_k}}{\mathbf{x}_{\Pi}} = \frac{\mathbf{x}_{d_k}}{\sum_{d_k \in \Pi} \mathbf{x}_{d_k}} \quad (45)$$

$$\text{Allocation weight } (d_k | k \in \{1, 2, 3, 4\}, s) = \omega'_{d_k} = \frac{s_{d_k}}{s_{\Pi}} = \frac{s_{d_k}}{\sum_{d_k \in \Pi} s_{d_k}} \quad (46)$$

Thus it is possible to measure for any portfolio holding entity their concentration of contributions in impact (eq. (45)) or effort (eq. (46)) in each academic domain $d_1 \in D_1$, department $d_2 \in D_2$, discipline $d_3 \in D_3$, and direction $d_4 \in D_4$.

6.9 Diversification Ratio

The diversification ratio (DR) is a measure of how much diversification benefits a portfolio by comparing the weighted average volatility of the individual assets to the actual risk of the portfolio. One expects that if fields of science are not perfectly correlated in their average returns, that diversification would yield some benefit measurable by this ratio.

$$\text{Diversification ratio} = \frac{\sum_{m \in \Pi} \omega_m \sigma_m}{\sigma_{\Pi}} \quad (47)$$

This ratio is lowerbounded by 1, which is the case that all manuscripts are perfectly correlated in their returns. This is likely only possible for small portfolios that have works within a very narrow niche, such that all works are likely to be cited by new works in that narrow niche. The larger the DR, the more the diversification from existing assets in the portfolio is benefiting the portfolio in terms of reducing volatility. This measure could be used to quantify the volatility of scientific careers and reveal the effects of risk for different degrees interdisciplinary collaboration. Together with the allocation concentration metrics, it is possible to reveal patterns in scales of success and impact for varying levels of interdisciplinary academic research work.

6.10 Funding and Time Research Efficiency

Two useful ways to measure how efficiently research is conducted is to look at the amount of academic capital $\mathbb{X}$ generated for the time and/or money invested into the research. Funding research efficiency measures how much $\mathbb{X}$ each dollar (or other currency or choice) of research funding generates on a given portfolio Π .

$$\text{Funding Efficiency} = \epsilon_{\$, \Pi} = \frac{\mathbb{X}_{\Pi}}{\$_{\Pi}} \quad (48)$$

This can be measured at the global level, which would quantify how much academic impact taxpayer funding is generating across the world, but can also be calculated for more granular portfolios. In cases where an academic work has multiple sources of funding, Liberata will ask the uploading author to define the proportional share of funding of each source contributed to the work to avoid double counting. If this information is not given, Liberata scales the funding tied to each author by their contribution share. If any single author has multiple funds tied to them, Liberata will take the simple average of the funding to figure subdivide the capital attribution to each source of funding.

Time research efficiency measures how much $\mathbb{X}$ is generated per unit time for a given entity's (i.e. individual, institution, geographic region, academic field, etc.) portfolio Π .

$$\text{Time Efficiency} = \epsilon_{t, \Pi} = \frac{\mathbb{X}_{\Pi}}{\Delta t_{\Pi}} \quad (49)$$

For each paper uploaded onto Liberata, the uploading author is asked to indicate when the work began. Δt_{Π} is the time from the beginning of the earliest work in the portfolio to the upload date of the latest work.

These metrics are intended to quantify how effectively academic entities can convert time and funding into academic impact, and depending on choice of portfolio, can have different yet meaningful interpretations.

6.11 Peer Reviewer and Replicator Reliability

One way of splitting up a portfolio for an individual contributor is by their contribution role (author, peer reviewer, replicator). On Liberata, if any work is retracted, the value of shares on that work automatically go to zero. Liberata discerns the individual's reliability from two metrics. The first is the proportional loss metric, which is defined below.

$$\text{Proportional Loss} = \%_L = \frac{\mathbb{X}_{lost}}{\mathbb{X}_{remaining}} \quad (50)$$

This quantity for measuring a peer reviewer's reliability would have that individual's total academic capital from peer review lost in the numerator, and total academic capital from peer review remaining in the denominator. Likewise the proportional loss for replicators are computed the same way, just on a replication academic capital. This quantity is meant to capture the tendency to not catch serious errors in reviewed or replicated works relative to tendency to catch them.

The second number is the proportional split metric, which is defined below.

$$\text{Proportional Split} = \%_S = \frac{\mathbf{x}_{task}}{\mathbf{x}_{total}} \quad (51)$$

This quantity for measuring a peer reviewer's reliability would have that individual's academic capital received from peer review in the numerator, and their total academic capital in the denominator. Likewise, replicator reliability is calculated the same way, just with academic capital gained from replication in the numerator. This quantity is meant to capture how frequently a reviewer or replicator is doing original research versus quality control.

These two numbers are not combined but instead both available for view on the Liberata platform for marketplace goers and readers to better judge the reliability of quality control done. In addition, documents on Liberata are version controlled, allowing users to read the peer reviewer comments as well as the manuscript prior and post each revision. Replications are written up and attached to the document under the automatically added replications section. An important difference is that on Liberata, peer reviewers are still anonymous, but replicators are known to authors and readers.

6.12 Impact of Quality Control

For any manuscript m , we can fit curves to the amount of returns $\mathbf{x}'_m$ it acquires over time t . The returns are expected to follow continuous and differentiable trajectories under normal circumstances, but may become non-differentiable due to the impact of a quality control event, among other things. Let's say for a Δt interval surrounding around a quality control event at time t , the returns changes at t as $\Delta \frac{d}{dt} \mathbf{x}_m(\Delta t)$, and in the same interval, the returns of the field d_4 changed by $\Delta \frac{d}{dt} \mathbf{x}_{d4}(\Delta t)$ then we compute the impact as the change in slope normalized by the change in the slope for the field.

$$\text{IQC} = \frac{\Delta \frac{d}{dt} \mathbf{x}_m(\Delta t)}{\Delta \frac{d}{dt} \mathbf{x}_{d4}(\Delta t)} = \frac{\frac{d}{dt} \mathbf{x}_m(t + \Delta t/2) - \frac{d}{dt} \mathbf{x}_m(t - \Delta t/2)}{\frac{d}{dt} \mathbf{x}_{d4}(t + \Delta t/2) - \frac{d}{dt} \mathbf{x}_{d4}(t - \Delta t/2)} \quad (52)$$

Then, if the quality control event (completed reviews or replications) helped in making the case in the article much stronger, then the IQC would be large. Thus, this value can be used to attribute the impact of the quality control activity to the QC service providers (i.e., reviewers and replicators). Using the shares owned by the reviewers/replicators on the manuscript m , the quantity of impact can also be suitably attributed to the contribution of each reviewer/replicator.

6.13 Collections & Journals

Many traditional scientometrics are well defined for only individuals. For example, citation counts and H-index are easy to compute for individuals, but not well defined for institutions, geographic regions, time periods, or academic fields. If a manuscript has two authors from institution A, and one from institution B, it is not clear for citation count or H-index if this manuscript should be counted once or twice for institution A, with both having problematic implications for A's credit relative to B.

In the Liberata system however, the notion of portfolios extend naturally to any set of shares. Portfolios need not be for only individuals. This allows powerful ways of comparing many different types of entities, and also enables a much simpler and consistent way of defining journals. In the Liberata system, journals are defined by collections of D_4 tags. Any manuscript with at least those tags automatically is populated into the journal. (Other filter tags exist for peer reviewed, replicated, etc.) For a journal with manuscripts M_J and total academic capital $\mathbf{x}_J$, the impact factor of a journal is replaced with the average academic capital $\mathbb{E}(m_J)$ of the manuscripts in the journal.

$$\mathbb{E}(m_J) = \frac{\mathbf{x}_J}{|M_J|} \quad (53)$$

This measure can be used to measure average impactfulness of institutions, geographic regions, academic fields, and intersections of these and more, by arbitrary selection of which manuscripts to include in the collection. All other metrics in this section also are applicable to any collection, which are ultimately just a type of portfolio.

7 Market Metrics

7.1 Fair Market Prices

For a transaction to occur on the Liberata platform, both the author and the peer reviewer or replicator must have positive expected value from its execution. As more transactions accumulate on Liberata marketplaces, a more confident value can be assessed for the fair market price (FMP) of peer review and replication services for each academic field. The definitions of the peer review FMP is:

$$\text{FMP}_{p,d} = \mathbb{E}[s_{p,d}] = \bar{s}_{p,d} \quad (54)$$

where $s_{p,d}$ denotes the shares that an average peer reviewer p in the academic field d holds on a work within d . Likewise, the definition of FMP for replication is:

$$\text{FMP}_{r,d} = \mathbb{E}[s_{r,d}] = \bar{s}_{r,d} \quad (55)$$

where $s_{r,d}$ denotes the shares that an average replicator r in the academic field d holds on a work within d . Note that we expect the two FMP values to be different for any d , because in general, peer review and replication take different amounts of effort to do. It is expected that for most fields, a successful replication will take more effort, but also be a much stronger signal for accuracy for any given work. Thus, we expect that $\text{FMP}_{r,d} > \text{FMP}_{p,d}$ in most cases, but this fundamentally does not always have to be true. It would be interesting to examine the fields where the reverse is true, such as a field where one can readily rerun simple code to check the author's results. In such fields, it may be possible to drive FMP_p to 0 unless the reviewers are able to add value to the work besides accuracy, such as clarity or interpretation nuance. The FMP values capture information about the perceived value add that peer review and replication has for a work by the members of that academic community who participate in the marketplace, as well as the risk of that work.

Section 6.3 also implies that we can have different 'depths' (domain, department, discipline, direction) of FMP with different values, for each of the four levels of tags. This provides a useful tradeoff between accuracy of the FMP measure, versus the simplicity of the computation.

7.2 Risk Premiums

Risk premium is the difference between the price (in shares) that a particular author or set of authors pay on average and the fair market price of the author(s)' academic field. The definition for risk premium for the peer review marketplace is:

$$\text{Risk premium}_{p,d} = \psi_{p,d}(\hat{A}) = \hat{s}_{p,d} - \bar{s}_{p,d} \quad (56)$$

where $\hat{s}_{p,d}$ denotes the average shares that author(s) $\hat{A}$ pay for peer review p within academic field d , and $\bar{s}_{p,d}$ is the FMP for peer review in that field. Likewise, the definition for risk premium for the replication marketplace is:

$$\text{Risk premium}_{r,d} = \psi_{r,d}(\hat{A}) = \hat{s}_{r,d} - \bar{s}_{r,d} \quad (57)$$

This quantity measures how much additional risk the community within d perceive for the author's work above the mean. (Note: this quantity can be negative if the author(s) $\hat{A}$ are seen as less risky than the average member of the field.) Higher risk premiums can indicate that the author(s) may (1.) have a poor track record of accurate work, (2.) are a newcomer to the field, (3.) have trouble clearly writing descriptions of their work, and/or (4.) are just doing work that is perceived to be less impactful. Whatever the reason, and their might be other reasons than the four identified here, peer reviewers and/or replicators would rather do quality control for other works in the field given the same share compensation and the additional incentive needed is useful information.

This quantity can be ascertained for any group of authors, such as all the researchers of an academic institution. The risk premiums paid by institutions can represent (1.) the track record of accurate work from that institution, (2.) the clarity of writing and presentation of work from that institution, (3.) the ease of working with authors from that institution, and (4.) perceived impactfulness of the work from that institution. This factors are not an exhaustive list but are or are among the expected major contributors to risk premium. Additionally, within an institution, risk can be calculated for different departments by sectioning authors and works according to their d_2 primary tags, allowing for the identification of particularly distinguished fields within the institution, i.e. the institution's specialty.